Cisco PE Node MAR-PE-13 QoS Congestion Event Resolution SOP

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_9595

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation on Cisco PE node MAR-PE-13 located at Marathahalli Core. The Cisco PE node is responsible for ensuring high-quality, low-latency traffic delivery between multiple network segments. QoS congestion events occur when the node's forwarding engine is overwhelmed by an excessive amount of traffic, causing interface congestion and subsequent packet drops. This can be due to various factors, such as misconfigured QoS policies, inadequate buffer sizes, or hardware limitations. In this SOP, we will investigate and resolve the root cause of the QoS congestion event, identified by alarm code ALARM_9595 and severity warning, to ensure reliable service delivery and prevent potential network outages.

Pre-requisite Checks

- prerequisite: Verify redundant link status on MAR-PE-13, description: Perform a show interface status command to confirm that all redundant links are up and functioning correctly.

- prerequisite: Check for any recent hardware or software changes on MAR-PE-13, description: Review recent maintenance logs, configuration changes, or software upgrades that may have impacted QoS performance.

- prerequisite: Ensure all QoS policies are properly configured on MAR-PE-13, description: Verify that all QoS policies, including class maps, policy maps, and service policies, are correctly applied and configured to prevent misbehaving traffic from congesting the network.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate buffer sizes	Medium
Hardware limitations	Low

Diagnostic Commands & Expected Outputs

```
Command: show queueing interface GigabitEthernet1/0/1, Expected Output: Interface GigabitEthernet1/0/1 is up and queueing is enabled, Description: This command displays the queueing configuration and statistics for the specified interface. We are looking for any signs of congestion, such as high packet drops or queue depths.
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```
Command: show queueing policy-map GigabitEthernet1/0/1, Expected Output: Policy map GigabitEthernet1/0/1 is applied and configured correctly, Description: This command displays the QoS policy configuration for the specified interface. We are looking for any misconfigured policies that may be causing congestion.

Command: show buffer, Expected Output: Buffer sizes are adequate and not causing congestion, Description: This command displays buffer sizes and usage statistics for the node. We are looking for any indications of buffer congestion that may be causing packet drops.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: show running-config | include class-map, Description: Review the QoS policy configuration to ensure that all class maps, policy maps, and service policies are correctly applied and configured to prevent misbehaving traffic from congesting the network. Make any necessary corrections and reapply the QoS policy.

Step 2: Step: Increase buffer sizes, Command: buffer-size 10000, Description: Increase the buffer sizes to prevent congestion and packet drops. This may require adjusting the buffer allocation to ensure adequate space for high-priority traffic.

Step 3: Step: Upgrade hardware, Command: upgrade hw, Description: If the root cause is hardware limitations, consider upgrading the node's hardware to improve QoS performance and reduce congestion.

Post-Check Verification

```
Command: show queueing interface GigabitEthernet1/0/1, Expected Output: Interface GigabitEthernet1/0/1 is up and queueing is enabled with no signs of congestion, Description: Verify that the QoS congestion event has been resolved by checking the interface queueing configuration and statistics.

Command: show queueing policy-map GigabitEthernet1/0/1, Expected Output: Policy map GigabitEthernet1/0/1 is applied and configured correctly, Description: Verify that the QoS policy configuration has been corrected and is functioning correctly.

Command: show buffer, Expected Output: Buffer sizes are adequate and not causing congestion, Description: Verify that the buffer sizes have been increased and are not causing congestion.
```

Roll-back Procedures

- Step: Revert QoS policy configuration, Command: no class-map, Description: Revert the QoS policy configuration to its previous state if necessary.

- Step: Decrease buffer sizes, Command: buffer-size 5000, Description: Decrease the buffer sizes to their previous state if necessary.

- Step: Downgrade hardware, Command: downgrade hw, Description: Downgrade the node's hardware to its previous state if necessary.